Architecture Decision Record: ADR-002

Implementing Pydantic Logfire Structured Logging

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Status: Accepted
Project: DVerse Communication Platform 2.0

1 Context

The DVerse platform requires structured visibility into the behavior of its AI agents. As agents communicate over Zenoh and through the chat application, ordinary freeform logs are insufficient for debugging and analysis. Engineers need logs that can be searched, filtered, correlated with traces, and validated against consistent schemas.

The agent layer is implemented in Python and uses Pydantic models. This makes Pydantic Logfire a strong fit for structured, schema-aware logging because it can work naturally with the models already used by the system.

2 Decision

Sprint 4 will implement Pydantic Logfire for structured logging across the agent layer. Logs will be emitted at major system boundaries and will be designed to correlate with OpenTelemetry traces.

The implementation will add structured logs at the following boundaries:

- Agent initialization and configuration loading.
- Incoming and outgoing Zenoh messages.
- Chat application events involving agents.
- Validation failures for agent messages or payloads.
- Runtime exceptions and recoverable error states.

Logs should include stable fields such as agent identifier, topic name, message identifier, trace identifier, timestamp, event type, and validation status. This will make logs queryable, consistent, and easier to correlate with OpenTelemetry traces.

3 Alternative Decisions

One alternative was to rely on plain JSON logs. This was rejected because plain JSON logging would require more manual serialization and would not provide the same Pydantic-native ergonomics.

Another alternative was to defer structured logging until after the communication layer was complete. This was rejected because logs are most useful when they are designed alongside the events and schemas they describe.

4 Consequences

4.1 Positive

- Logs will be structured, queryable, and consistent across major agent boundaries.
- Pydantic model awareness will reduce manual serialization work.
- Logs can be correlated with OpenTelemetry traces through shared trace identifiers.

4.2 Negative

- The team must define and maintain stable logging schemas.
- Logging introduces additional operational setup and runtime overhead.
- Sensitive fields must be reviewed carefully before being emitted to the logging system.

5 Scope

Deliverable	**Description**
Agent logs	Emit structured logs for initialization, configuration, and lifecycle events.
Message logs	Log incoming and outgoing Zenoh messages with stable identifiers.
Chat logs	Log chat application events involving AI agents.
Error logs	Capture validation failures, exceptions, and recoverable error states.

Statement on AI Use

Artificial intelligence was used to support the drafting of parts of this document. The generated text was subsequently reviewed, checked, and edited where necessary by me, the author, Abel-Raul Mazilu.